BranchBox: Stage and Target Supervision for Multi-Object Placement

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Abstract

Multi-object manipulation requires a robot to switch both its active subtask and its spatial target. We study structured supervision with BranchBox, a vision-language-action training recipe combining stage-semantic instructions, a separate marked image, and auxiliary stage and target heads on action-decoder features. Prompt dropout and temporal perturbations support ordinary test inputs or prediction feedback through a fixed task-specific stage table, without an additional learned planner. In a post hoc exploratory analysis of four two-object LIBERO-Long tasks, SmolVLA and X-VLA use matched training data and paired initial states. With self-generated prompts, success reaches 56.9% versus 45.6% for matched action-only training: an 11.3-percentage-point gain (unadjusted 95% CI: 1.9 to 20.6) and a 24.7% relative increase. Both models have positive point estimates; ordinary test inputs yield 55.6% success. The positive evidence concerns the combined training recipe within this selected task family. Complete broader-study results and separate memory-extension experiments document its scope and unresolved limitations; generality beyond the selected tasks, models and training seed remains to be established.

1. Introduction

Placing two objects requires more than repeating a single grasp. After completing one placement, the robot must select the next object, shift its spatial target, and continue toward the remaining goal. A task instruction specifies what should be achieved, but a vision-language-action (VLA) policy must infer this changing stage and target from its observations [3, 12, 20, 28]. We ask whether a compact, supervised stage/target interface can improve these routines when no additional learned planner runs online.

Intermediate representations are already useful interfaces for robot learning. RT-H uses language motions [2], ECoT predicts semantic and visually grounded quantities [25], VP-VLA conditions a controller on stage commands and marked images [21], and GuidedVLA supervises task-relevant factors in action-decoder attention [11]. We build on this line of work with a compact implementation for lightweight policies: two small heads predict a stage and spatial target from action-decoder features, while training also exposes the controller to stage text and visual marks.

BranchBox uses a separate prompt-image slot so that the original camera views remain available. Text and marks are independently dropped and temporally perturbed during training. The trained network can then act from ordinary observations, or return its previous predictions through the text and image interface. A fixed task-specific table maps the predicted stage to its instruction and mark type; deployment does not require an additional learned planner or tracker. This table supplies task structure, while the policy predicts where it is within that structure.

We study four two-object placement tasks from our evaluated LIBERO-Long set [15]: two moka pots, two mugs, a mug and pudding, and soup and sauce. In the focused exploratory analysis, success increases from 45.6% with matched action-only training to 56.9% with BranchBox and self-generated prompts: an absolute gain of 11.3 percentage points and a 24.7% relative increase. Both model families and all four tasks have positive point estimates. The same trained networks reach 55.6% with ordinary inputs, supporting deployment when test-time prompting is undesirable.

The improvement with ordinary inputs shows that the combined training recipe can improve the evaluated policy even when deployment supplies no generated cue. The feedback mode provides an additional interface for the same trained policy. These comparisons evaluate the combined recipe; they do not isolate the auxiliary heads from prompt

exposure or establish a separate gain from feedback. We also examine richer history in supplementary studies, keeping those interventions separate from the original recipe and its paired evaluation.

Contributions. We present (1) a structured training recipe that supervises stage and spatial-target prediction on action-decoder features while perturbing the corresponding input prompts; (2) a deployment interface that supports ordinary observations or self-generated prompts through a supplied task-specific stage table; and (3) a paired analysis across two lightweight policies, with positive point estimates for both deployment modes on four two-object tasks. We report counts, paired uncertainty and all task-level outcomes rather than selecting a deployment mode separately for each task. The supplementary material preserves the broader experimental scope and additional history interventions, including unsuccessful variants.

2. Related Work

Visual prompts at the policy input. Marks rendered into an image let multimodal models refer to regions: Set-of-Mark labels segments for GPT-4V [23], MOKA pairs task decomposition with marked keypoints for manipulation [5], and PIVOT iteratively draws candidate actions for a vision-language model to select [16]. TraceVLA [29] uses motion traces from a point tracker. Policies conditioned on drawn trajectories (RT-Trajectory [6], HAMSTER [13]) or pretrained with mark supervision (Magma [24]) learn to read the cue during training. VP-VLA [21] uses a Qwen3-VL planner [1] and SAM3 grounding to construct stage instructions and visual prompts. Its auxiliary grounding objective predicts discretized coordinates at key frames using cross-entropy, with grounding gradients updating the VLM. We retain its separate prompt-image interface, while stage classification and continuous coordinate regression supervise the action decoder with the visual/language backbones frozen in the original two-model study.

Hierarchical language conditioning. SayCan combines a language model's estimate of skill relevance with learned affordances [10], and Inner Monologue feeds success detection and scene descriptions back to a language planner [9]. RT-H trains a policy to predict intermediate language motions before actions [2]; YAY Robot uses human language corrections to a high-level policy [18]; Hi Robot trains a vision-language model to issue commands to a VLA [19]; and $\pi_{0.5}$ predicts a semantic subtask before its actions [3]. BranchBox shares the use of an explicit intermediate representation, and pairs a stage with a spatial target, inside one lightweight network and with no second model at test time.

Policy-generated intermediate representations. ECoT [25] trains a VLA to generate task decompositions and visually grounded quantities, including object boxes, before predicting actions. CoT-VLA [27] predicts a future image as an intermediate visual goal, while DiffusionVLA [22] injects self-generated language reasoning into its action model. SPR [4] predicts remaining spatial subtasks and gripper trajectories, and monitors progress to trigger a learned rewind behavior. These methods predict structured intermediate representations within the policy. GuidedVLA [11] supervises designated action-decoder attention heads for object and skill factors, with depth features and residual attention fusion. BranchBox instead uses small heads on pooled action-decoder features to predict a stage index and spatial target. A supplied task-specific stage table converts these predictions into delayed text and image prompts. This avoids autoregressive generation of a reasoning sequence or goal image, but does not provide open-ended task decomposition. We evaluate the training recipe separately from the effect of feeding its predictions back into the policy.

Lightweight VLAs. We instantiate the recipe in SmolVLA [20] and X-VLA [28], two compact policies with different action decoders. The original two-model recipe trains the decoder and its auxiliary heads while freezing the visual and language backbones. The architecture also includes a prompt-image slot; we do not claim a measured system-cost advantage over the above methods. The separate memory extensions explore both full-weight and parameter-efficient adaptation.

Interventions around frozen policies. BYOVLA edits task-irrelevant image regions at run time [8], RoboHarness adds episodic memory and failure attribution [14], and Harness VLA composes a frozen VLA with analytic primitives [26]. These methods leave the weights alone. Our work instead studies structured supervision during policy training.

Recurrent policy memory. MemBodied [17] combines a recurrent associative state with an initial-scene anchor. SmoLSTM [7] uses persistent matrix-LSTM state and evaluates the trained policy with its state reset between decisions. Our additional X-VLA studies adopt MemBodied's core mechanisms and use same-weight state/anchor interventions to test reliance on history. Their contrasts with the existing checkpoint also include additional training; it is not an isolated comparison of memory architectures or a reproduction of the papers' published benchmark scores.

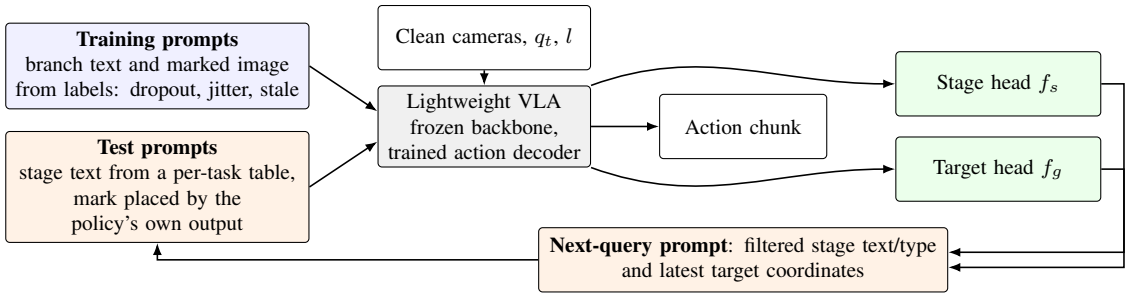


Figure 1. BranchBox in the original two-model study. In training, labels supply the prompts, and the two heads learn the current stage and target. At test time the previous query's predictions supply the next prompt through a fixed task table. Stage predictions are filtered; target coordinates are taken from the latest query. No external planner, tracker or simulator state supplies these predictions.

3. Method

Task interface and action policy. At query t, a policy receives camera images I_t, proprioception q_t and instruction l, and predicts an action chunk A_t. The first h actions are executed before the next query. Each task has an ordered table $\mathcal{T} = \{(s_k, m_k)\}_{k=0}^{K-1}$, where s_k is stage text and m_k is a crosshair, box, or no mark. We fix the table from demonstration annotation metadata before training and supply it at test time. The policy predicts progress within this supplied decomposition. Our implementation supports at most six stages.

Training prompts and supervision. The branch instruction appends `Current subtask:` and s_{k_t} to l. Following VP-VLA [21], the main camera stays clean and a marked copy enters a separate image slot; the wrist camera is not marked. A crosshair denotes an object to reach, and a box denotes a destination or part. Offline simulator predicates, contacts and segmentation provide training-stage and normalized image-target annotations. For the two-object tasks, k_t is twice the subtask position plus a reach/place phase, giving four stages (Tab. 1). The two phases share a subtask text but use different mark types. The supplementary material specifies the annotation rules; the test-time prompter never queries these labels.

Prompt perturbations. Text and marks are independently retained with probability 0.5. Mark coordinates are independently perturbed by up to 5% of the image extent and clipped to $[0, 1]$. At demonstration frame u, with probability 0.5, prompt labels come from frame $u - L$, with L uniform on $1, \ldots, 3h$. Otherwise, a frame offset in $\{-5, \ldots, -1, 1, \ldots, 5\}$ is applied with probability 0.5. Indices are clipped to the current demonstration. Labels from the selected frame are rendered onto the current image, while auxiliary supervision always uses the current frame. Draws are deterministic functions of the seed, sample index and epoch. These perturbations expose the network to missing and delayed cues while retaining current-frame supervision.

Auxiliary prediction. Two MLPs, each with two 256-unit ReLU hidden layers, read the input features to the action projection, averaged over action-chunk tokens. The supplementary material gives the model-specific attachment points. The stage head has six logits; the target head outputs $(p_x, p_y, x_0, y_0, x_1, y_1)$ through a sigmoid in normalized image coordinates. The loss is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + 0.1\mathcal{L}_{\text{stage}} + 0.1\mathcal{L}_{\text{target}}. \tag{1}$$

Stage loss is six-way cross-entropy on labeled samples; task-specific class restriction is used only at deployment. Target loss applies L1 supervision only to valid coordinates of the annotated mark type: point coordinates for crosshairs and box coordinates for boxes. Within each valid sample the absolute errors are averaged over its two or four supervised coordinates, then averaged over valid samples; a batch with no valid targets has zero target loss. Thus, not all six target outputs are supervised on every frame. Both auxiliary weights are fixed at 0.1; the objectives are stage cross-entropy and continuous target regression, distinct from VP-VLA's grounding objective. In the original two-model recipe, the visual and language backbones remain frozen, while auxiliary gradients update the action-decoder representation. Training features come from the training forward pass, while inference features are taken from the final action-generation step.

Feedback at test time. The first query uses stage-0 text and a clean prompt image. After each new chunk, we take the stage argmax only over the K stages valid for the task ($K = 4$ in the focused tasks). The filtered stage never decreases and advances after two consecutive predictions agree on a later stage; jumps over intermediate stages are allowed. The next query uses the filtered stage's text and mark type, drawing the latest target coordinates on the new

camera image without tracking or geometric warping. Only observations, policy outputs and the fixed task table enter this loop.

The filter and target are updated on different rules: when their stages disagree, a filtered-stage mark can use coordinates not supervised for that mark type on the latest frame. The reported results retain this original feedback rule. The supplementary material documents this limitation and a separate aligned-controller caching pilot; the pilot is not substituted for the original PS arm.

Persistent-history extension. As a separate extension, we adapt MemBodied [17] to X-VLA: associative reads and execution-conditioned writes in all 24 action-expert layers, dedicated state/memory tokens, and a first-frame episode anchor. These additions are absent from the original BranchBox results. The supplementary material specifies the backbone mapping, training attempts and same-weight state/anchor interventions.

4. Evaluation Protocol

The main study evaluates the original two-model training recipe. Additional memory studies use different checkpoints or controllers and different initial-state panels; their outcomes are analyzed separately.

Focused task family. Among the six evaluated LIBERO-Long tasks, four require placing two objects. We examine these four tasks and two model families as a retrospectively selected exploratory subset. The tasks share a four-stage reach/place structure (Tab. 1, Fig. 2). The complete broader study, including the remaining task types, model configurations, original primary analysis and protocol deviations, is retained in the supplementary material. This focus is not a prespecified endpoint or a benchmark-wide performance claim.

Original two-model training. Training uses 203 demonstrations and 58,272 frames from all six evaluated LIBERO tasks, including the two tasks outside the focused analysis. Thus, the present grouping changes the analysis scope, not the trained checkpoints or their data. SmolVLA [20] starts from `HuggingFaceVLA/smolvla_libero` and trains its action expert and state/action projections for 14,000 steps. X-VLA [28] starts from `lerobot/xvla-base` and trains its transformer and soft prompts for 20,000 steps. Batch size is 32 and the training seed is 1. Native and BranchBox runs share initial weights, sample order, update counts and seed; BranchBox adds prompt exposure and auxiliary supervision. We evaluate only final checkpoints. The supplementary material records demonstration filtering, label construction, adapter settings, and feedback pseudocode. Demonstrations are replayed from their recorded initial states; evaluation uses the official initial-state list.

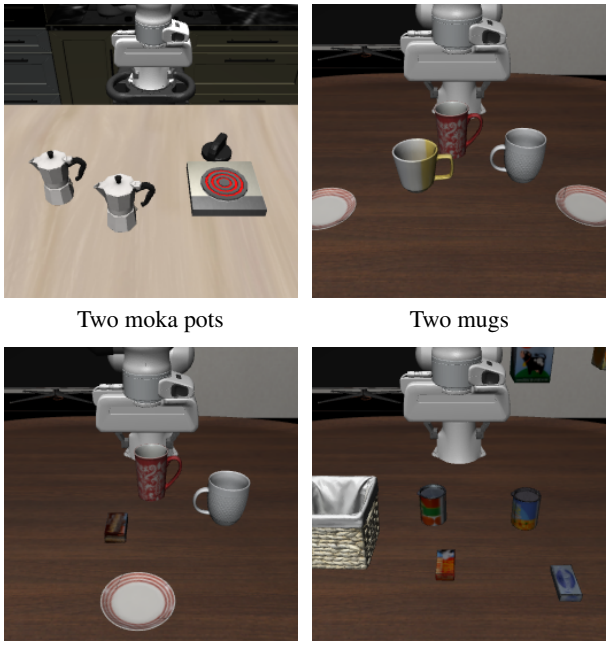


Figure 2. The four two-object placement routines. These unmodified main-camera observations show official initial-state 40 from the separate development panel, before any policy action. The same fixed state index was chosen for each task before memory evaluation; these are task illustrations, not selected success trajectories.

Matched evaluation arms. N0 is the matched action-only baseline with the plain instruction and a clean main-camera copy in the prompt slot. P0 is the BranchBox-trained network with the same inputs. PS uses that network with its own generated prompts. All arms contain the same image slots and head modules; N0 auxiliary heads have zero training weight. N0 therefore controls the recipe within this shared architecture; it is not the untouched official checkpoint or its published benchmark score. Comparisons against N0 measure the combined training recipe under each deployment mode. P0 uses no external prompt source, and PS uses only policy predictions and its fixed task table.

Original paired evaluation. Each model–task–arm runs official initial states 20–39 with matched episode seeds. The four tasks provide 80 distinct task–initial-state pairs, each evaluated by both models: 160 model–scene evaluations per arm and 480 episodes in total. Initial-frame hashes and episode seeds match across all six model–arm combinations on every focused scene. Images are 256×256; ten settling

Table 1. Supplied decompositions for the focused LIBERO-Long tasks (zero-based `libero_10` IDs). Each subtask uses two stages with the same text: a crosshair on its object, then a box on its destination, giving four stages per task. Exact strings are in the supplementary material.

Task (ID)	First subtask	Second subtask
Moka pots (8)	Second pot → stove	First pot → stove
Two mugs (4)	White mug → left plate	Yellow/white mug → right plate
Mug/pudding (6)	White mug → plate	Pudding → right of plate
Soup/sauce (0)	Alphabet soup → basket	Tomato sauce → basket

actions precede a 520-step horizon. SmolVLA executes 10 actions per query and X-VLA executes 30.

For each bootstrap replicate, we independently resample the 20 initial states within each task and share those indices across *both models and all arms*. We then average equally over tasks and models. This preserves their common-scene dependence. We use $B = 100{,}000$ replicates, seed 20261101, and percentile intervals. Intervals condition on the fixed checkpoints and tasks; they are unadjusted for retrospective subset selection and multiple contrasts. The supplementary material gives the exact task order, a resampling sensitivity check, and complete broader-study results. The source package includes per-episode success indicators and an executable analysis script.

Separate memory studies. Three additional X-VLA studies examine a target cache (40 pairs), a final 6,000-update full-weight memory adaptation (20 triples), and a validation-selected LoRA memory adaptation (180 quadruples). All use training and rollout seed one, and every panel includes previously used development scenes. The expanded study adds a state-only reset that retains the first-frame anchor, separating its contrast from the joint state/anchor reset. The supplementary material reports the protocols, selection history, all comparisons and diagnostics. These interventions are not extra samples for the original recipe comparison, and no equal-budget memory-free retraining control was run.

5. Results on Multi-Object Placement

An 11.3-point absolute gain and 24.7% relative increase. Across the four tasks and two models, PS solves 91 of 160 episodes compared with 73 for N0: 56.9% versus 45.6%, a difference of $+11.3$ percentage points (unadjusted 95% CI $+1.9$ to $+20.6$; Tab. 2). Among discordant pairs, PS succeeds on 38 model–scene pairs where N0 fails, versus 20 in the opposite direction. This corresponds to a 24.7% relative increase, computed from the unrounded success counts. This is the same effect expressed relative to matched action-only success, within the selected multi-object task family.

Positive point estimates in both model families. SmolVLA increases from 42/80 to 50/80 successes (52.5% to 62.5%; $+10.0$ points). X-VLA increases from 31/80 to 41/80 (38.8% to 51.3%; $+12.5$ points). Table 2 reports the wider per-model intervals. The common direction across the two architectures motivates further tests of the same task structure with independently trained policies.

A 10-point gain using ordinary test inputs. P0 solves 89/160 episodes (55.6%), compared with 73/160 for N0, for $+10.0$ points (unadjusted 95% CI 0.0 to $+19.4$). Its model-level differences are $+11.3$ points for SmolVLA and $+8.8$ for X-VLA. This deployment option retains the trained action policy while using a plain instruction and clean images at test time. The comparison measures the full training recipe, including both auxiliary supervision and prompt exposure. P0 and PS are deployment choices for the same trained policy; their direct comparison appears in the supplementary material.

Gains span four distinct two-object routines. All four tasks have positive PS$-$N0 and P0$-$N0 pooled point estimates (Tab. 3). With self-generated prompts, the differences are $+10.0$ points on moka pots, $+15.0$ on two mugs, $+7.5$ on mug and pudding, and $+12.5$ on soup and sauce. The largest P0 difference occurs on moka pots: 33/40 versus 25/40, or $+20.0$ points. Each deployment arm is reported separately rather than choosing the better arm for each task. These per-task estimates describe variation within the selected family; their uncertainty is shown in Fig. 3.

Interpreting the task structure. Each focused task requires two reach/place cycles and a change of object identity between them. Stage and target supervision gives the action decoder explicit labels for these intermediate variables. The observed pattern is compatible with this motivation, but successful stage transitions and target switching were not measured as causal mediators. Their contribution requires dedicated diagnostics rather than inference from task success alone.

5.1. Additional history: no established gain

The supplementary studies test whether richer history adds to the trained recipe. A one-query target cache solves 14/40 scenes versus 19/40 with reads disabled (-12.5 points; 95% CI $[-27.5, +2.5]$). A 6,000-update full-weight memory adaptation solves 12/20, versus 13/20 for the existing policy and 11/20 with same-weight joint state/anchor reset; both paired intervals include zero.

Table 2. Four two-object LIBERO placement tasks. The two models share 80 task–initial-state pairs. Brackets give unadjusted 95% paired bootstrap intervals, with scene indices shared across models and arms. The subset was selected retrospectively; complete broader-study comparisons are in the supplementary material.

Model	N0 matched	P0 ordinary input	PS self-prompt	P0−N0 (points)	PS−N0 (points)
SmolVLA	42/80 (52.5%)	51/80 (63.8%)	50/80 (62.5%)	+11.3 [-1.3, +23.8]	+10.0 [-2.5, +22.5]
X-VLA	31/80 (38.8%)	38/80 (47.5%)	41/80 (51.3%)	+8.8 [-3.8, +21.3]	+12.5 [-1.3, +26.3]
Both models	73/160 (45.6%)	89/160 (55.6%)	91/160 (56.9%)	+10.0 [0.0, +19.4]	+11.3 [+1.9, +20.6]

Table 3. Per-task success pooled over both models. Differences are percentage points; paired uncertainty is plotted in Fig. 3.

Task	N0	P0	PS	P0−N0	PS−N0
Moka pots	25/40	33/40	29/40	+20.0	+10.0
Two mugs	15/40	16/40	21/40	+2.5	+15.0
Mug and pudding	18/40	22/40	21/40	+10.0	+7.5
Soup and sauce	15/40	18/40	20/40	+7.5	+12.5

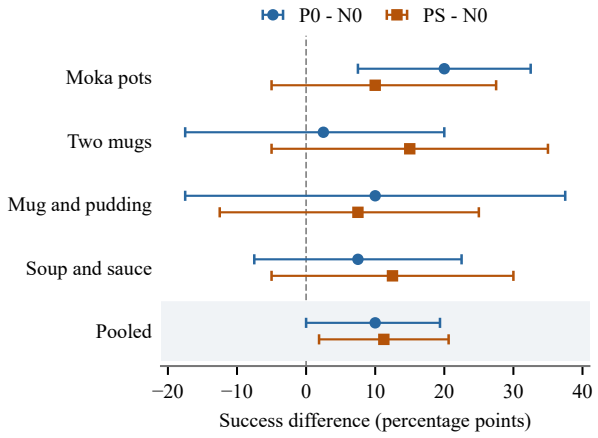


Figure 3. Differences against N0 on the selected two-object tasks. Points are observed effects; bars are unadjusted 95% scene-clustered paired bootstrap intervals. P0 and PS are fixed deployment modes, each evaluated on every task.

A larger follow-up evaluates all six tasks on 180 paired scenes with four conditions. Validation-selected LoRA memory solves 103/180, versus 107/180 when only recurrent state is reset, 113/180 under joint state/anchor reset, and 118/180 for the original policy. Its primary history contrast is -2.2 points (95% CI $[-8.9, +4.4]$; Holm-adjusted exact McNemar $p = 0.783$). Disabling reads barely changes the selected checkpoint's validation action loss. These findings do not establish an added memory benefit under the tested training conditions; they do not show that memory is ineffective in general. All comparisons, task-level increases and decreases, and training diagnostics are retained in the supplementary material.

6. Scope and Next Steps

The analysis concerns a task and model subset chosen after observing a broader experiment. Its intervals are conditional on one training seed and are not selection-adjusted, so the numerical gains should be interpreted as exploratory evidence for this task family. Independent seed pairs and independently selected multi-object tasks are the next tests of generality. The supplementary material preserves the complete original experimental scope, comparisons between deployment modes and implementation diagnostics.

The training recipe combines input prompts and auxiliary supervision; a heads-only training control would help identify their individual roles. The current interface also requires a supplied stage table and uses delayed predictions. Stage-consistent target feedback and recovery from incorrect stage transitions are concrete extensions to evaluate. These are future evaluation conditions, not changes underlying the reported results.

The memory extensions include separate recurrent-state and anchor interventions, but neither an equal-budget memory-free training control nor a fully independent evaluation panel. Worsening monitoring losses and weak sensitivity of the selected checkpoint to memory readout limit their interpretation. Neither the original recipe study nor these extensions establish transfer to new task decompositions or real robots.

7. Conclusion

BranchBox combines stage and spatial-target supervision on action-decoder features, with deployment through ordinary inputs or self-generated prompts. On a post hoc subset of four two-object LIBERO placement tasks, success rises from 45.6% to 56.9% with self-generated prompts: an absolute gain of 11.3 points and a 24.7% relative increase. Ordinary inputs reach 55.6%. Both model families and all four tasks have positive point estimates against matched action-only training. The additional memory variants do not establish a further gain under their tested conditions. These conditional gains motivate independent replication on new tasks and training seeds.

LLM Usage Statement

LLM-based assistants (OpenAI Codex and Anthropic Claude Code) supported research planning and literature retrieval, software implementation, debugging and testing, experiment operation on the cluster, analysis scripting, figure layout and plotting code, and manuscript drafting and editing under author direction. Experimental numbers were computed from the recorded episode and training logs. The authors are responsible for all claims.

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